

CS280A — Project 3

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A.1: Shoot and digitize pictures

Here I display three sets of photos, two of which are buildings and one of which is a wall.

Image 1 with points



View from the 6th floor of the Sutardja Dai Hall.

Image 2 with points



Image 1 with points



View from the skybridge between the Sutardja Dai Hall and Cory Hall.

Image 2 with points



Image 1 with points



Image 2 with points



Photo of a wall.

A.2: Building 1

System equations

From $\mathbf{p}' \sim H\mathbf{p}$ to a linear system $A\mathbf{h} = \mathbf{b}$.

$$\mathbf{p}' \sim H\mathbf{p}, \quad \mathbf{p} = [x, y, 1]^T, \quad \mathbf{p}' = [x', y', 1]^T$$

If we can fix $h_{33} = 1$, the equation above becomes

$$\begin{aligned} -x h_{11} - y h_{12} - h_{13} + x x' h_{31} + y x' h_{32} &= -x' \\ -x h_{21} - y h_{22} - h_{23} + x y' h_{31} + y y' h_{32} &= -y' \end{aligned}$$

where we denote $\mathbf{h} = [h_{11}, h_{12}, h_{13}, h_{21}, h_{22}, h_{23}, h_{31}, h_{32}]^T$, then we have $A\mathbf{h} = \mathbf{b}$.

Our numeric system for the building 1 (20 point correspondences in total)

Matrix A (shape 40x8):

-2138.000	-3400.000	-1.000	0.000	0.000	0.000	6751804.000	10737200.000
0.000	0.000	0.000	-2138.000	-3400.000	-1.000	7423136.000	11804800.000
-1914.000	-4040.000	-1.000	0.000	0.000	0.000	5592708.000	11804880.000
0.000	0.000	0.000	-1914.000	-4040.000	-1.000	7893336.000	16660960.000
-2022.000	-4028.000	-1.000	0.000	0.000	0.000	6134748.000	12220952.000
0.000	0.000	0.000	-2022.000	-4028.000	-1.000	8330640.000	16595360.000
-1914.000	-4268.000	-1.000	0.000	0.000	0.000	5585052.000	12454024.000
0.000	0.000	0.000	-1914.000	-4268.000	-1.000	8329728.000	18574336.000
-2014.000	-4256.000	-1.000	0.000	0.000	0.000	6086308.000	12861632.000
0.000	0.000	0.000	-2014.000	-4256.000	-1.000	8764928.000	18522112.000
-1902.000	-4548.000	-1.000	0.000	0.000	0.000	5534820.000	13234680.000
0.000	0.000	0.000	-1902.000	-4548.000	-1.000	8825280.000	21102720.000
-2006.000	-4540.000	-1.000	0.000	0.000	0.000	6054108.000	13701720.000
0.000	0.000	0.000	-2006.000	-4540.000	-1.000	9315864.000	21083760.000
-1898.000	-4768.000	-1.000	0.000	0.000	0.000	5515588.000	13855808.000
0.000	0.000	0.000	-1898.000	-4768.000	-1.000	9231872.000	23191552.000
-706.000	-3816.000	-1.000	0.000	0.000	0.000	1238324.000	6693264.000
0.000	0.000	0.000	-706.000	-3816.000	-1.000	2682800.000	14500800.000
-1230.000	-3784.000	-1.000	0.000	0.000	0.000	2747820.000	8453456.000
0.000	0.000	0.000	-1230.000	-3784.000	-1.000	4674000.000	14379200.000
-706.000	-4136.000	-1.000	0.000	0.000	0.000	1238324.000	7254544.000
0.000	0.000	0.000	-706.000	-4136.000	-1.000	2900248.000	16990688.000
-1222.000	-4104.000	-1.000	0.000	0.000	0.000	2725060.000	9151920.000
0.000	0.000	0.000	-1222.000	-4104.000	-1.000	5034640.000	16908480.000
-1218.000	-4328.000	-1.000	0.000	0.000	0.000	2711268.000	9634128.000
0.000	0.000	0.000	-1218.000	-4328.000	-1.000	5290992.000	18800832.000
-698.000	-4692.000	-1.000	0.000	0.000	0.000	1213124.000	8154696.000
0.000	0.000	0.000	-698.000	-4692.000	-1.000	3241512.000	21789648.000
-1210.000	-4632.000	-1.000	0.000	0.000	0.000	2678940.000	10255248.000
0.000	0.000	0.000	-1210.000	-4632.000	-1.000	5624080.000	21529536.000
-686.000	-4928.000	-1.000	0.000	0.000	0.000	1186780.000	8525440.000
0.000	0.000	0.000	-686.000	-4928.000	-1.000	3342192.000	24009216.000
-1206.000	-4860.000	-1.000	0.000	0.000	0.000	2670084.000	10760040.000
0.000	0.000	0.000	-1206.000	-4860.000	-1.000	5870808.000	23658480.000
-690.000	-5252.000	-1.000	0.000	0.000	0.000	1193700.000	9085960.000
0.000	0.000	0.000	-690.000	-5252.000	-1.000	3576960.000	27226368.000
-1194.000	-5172.000	-1.000	0.000	0.000	0.000	2624412.000	11368056.000
0.000	0.000	0.000	-1194.000	-5172.000	-1.000	6175368.000	26749584.000
-758.000	-3280.000	-1.000	0.000	0.000	0.000	1365916.000	5910560.000
0.000	0.000	0.000	-758.000	-3280.000	-1.000	2492304.000	10784640.000

Vector **b** (shape 40×1):

-3158.000
-3472.000
-2922.000
-4124.000
-3034.000
-4120.000
-2918.000
-4352.000
-3022.000
-4352.000
-2910.000
-4640.000
-3018.000
-4644.000
-2906.000
-4864.000
-1754.000
-3800.000
-2234.000
-3800.000
-1754.000
-4108.000
-2230.000
-4120.000
-2226.000
-4344.000
-1738.000
-4644.000
-2214.000
-4648.000
-1730.000
-4872.000
-2214.000
-4868.000
-1730.000
-5184.000
-2198.000
-5172.000
-1802.000
-3288.000

Recovered homography *H*

We report *H* normalized so that *H*₃₃ = 1:

0.775	-0.008	1157.522
-0.132	0.911	249.095
-0.000	-0.000	1.000



Point correspondences of the building 1.

A.2: Building 2

Our numeric system for the building 2 (10 point correspondences in total)

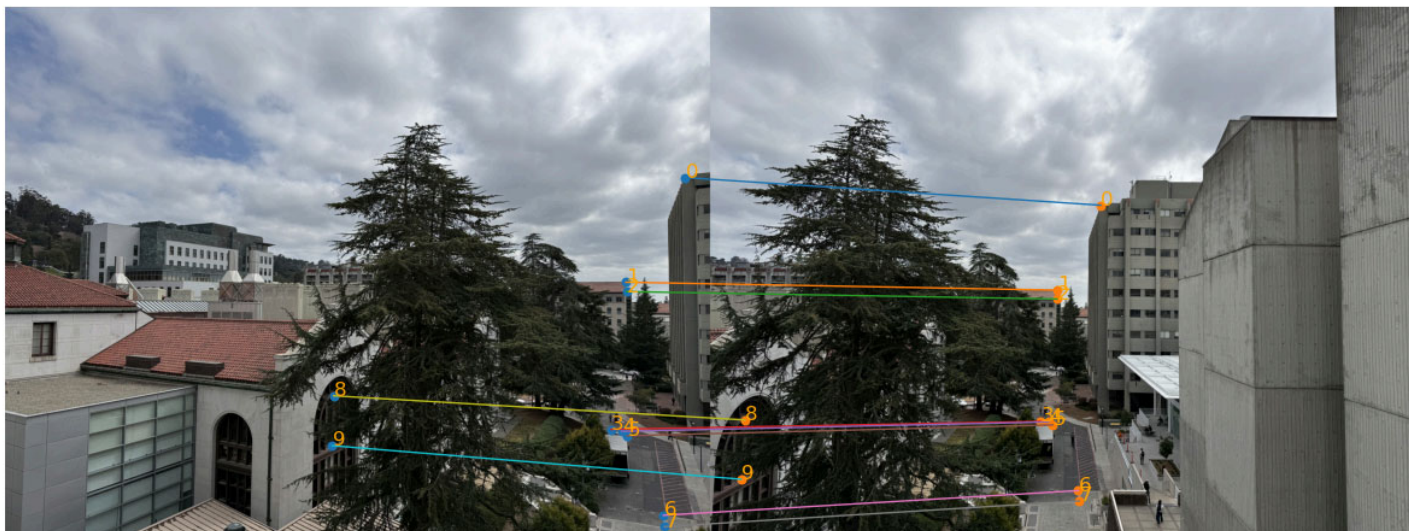
Matrix A (shape 20×8):

-5510.000	-1380.000	-1.000	0.000	0.000	0.000	17422620.000	4363560.000
0.000	0.000	0.000	-5510.000	-1380.000	-1.000	8838040.000	2213520.000
-5030.000	-2228.000	-1.000	0.000	0.000	0.000	14134300.000	6260680.000
0.000	0.000	0.000	-5030.000	-2228.000	-1.000	11528760.000	5106576.000
-5034.000	-2304.000	-1.000	0.000	0.000	0.000	14145540.000	6474240.000
0.000	0.000	0.000	-5034.000	-2304.000	-1.000	11900376.000	5446656.000
-4906.000	-3420.000	-1.000	0.000	0.000	0.000	13118644.000	9145080.000
0.000	0.000	0.000	-4906.000	-3420.000	-1.000	16464536.000	11477520.000
-4994.000	-3436.000	-1.000	0.000	0.000	0.000	13693548.000	9421512.000
0.000	0.000	0.000	-4994.000	-3436.000	-1.000	16779840.000	11544960.000
-5038.000	-3472.000	-1.000	0.000	0.000	0.000	13975412.000	9631328.000
0.000	0.000	0.000	-5038.000	-3472.000	-1.000	17048592.000	11749248.000
-5334.000	-4116.000	-1.000	0.000	0.000	0.000	15905988.000	12273912.000
0.000	0.000	0.000	-5334.000	-4116.000	-1.000	20887944.000	16118256.000
-5342.000	-4208.000	-1.000	0.000	0.000	0.000	15972580.000	12581920.000
0.000	0.000	0.000	-5342.000	-4208.000	-1.000	21368000.000	16832000.000
-2662.000	-3148.000	-1.000	0.000	0.000	0.000	750684.000	887736.000
0.000	0.000	0.000	-2662.000	-3148.000	-1.000	8901728.000	10526912.000
-2650.000	-3552.000	-1.000	0.000	0.000	0.000	662500.000	888000.000
0.000	0.000	0.000	-2650.000	-3552.000	-1.000	10133600.000	13582848.000

Vector b (shape 20×1):

$$\begin{bmatrix} -3162.000 \\ -1604.000 \\ -2810.000 \\ -2292.000 \\ -2810.000 \\ -2364.000 \\ -2674.000 \\ -3356.000 \\ -2742.000 \\ -3360.000 \\ -2774.000 \\ -3384.000 \\ -2982.000 \\ -3916.000 \\ -2990.000 \\ -4000.000 \\ -282.000 \\ -3344.000 \\ -250.000 \\ -3824.000 \end{bmatrix}$$
Recovered homography H

We report H normalized so that $H_{33} = 1$:

$$\begin{bmatrix} 2.274 & -0.116 & -5236.892 \\ 0.566 & 1.791 & -1975.948 \\ 0.000 & -0.000 & 1.000 \end{bmatrix}$$


Point correspondences of the building 2.

A.2: The wall

Our numeric system for the wall (15 point correspondences in total)

Matrix A (shape 30×8):

$$\begin{bmatrix} -4206.000 & -2080.000 & -1.000 & 0.000 & 0.000 & 0.000 & 8841012.000 & 4372160.000 \\ 0.000 & 0.000 & 0.000 & -4206.000 & -2080.000 & -1.000 & 8125992.000 & 4018560.000 \\ -3686.000 & -2104.000 & -1.000 & 0.000 & 0.000 & 0.000 & 6155620.000 & 3513680.000 \\ 0.000 & 0.000 & 0.000 & -3686.000 & -2104.000 & -1.000 & 7106608.000 & 4056512.000 \\ -4154.000 & -3460.000 & -1.000 & 0.000 & 0.000 & 0.000 & 8698476.000 & 7245240.000 \\ 0.000 & 0.000 & 0.000 & -4154.000 & -3460.000 & -1.000 & 13110024.000 & 10919760.000 \\ -3650.000 & -3404.000 & -1.000 & 0.000 & 0.000 & 0.000 & 6139300.000 & 5725528.000 \\ 0.000 & 0.000 & 0.000 & -3650.000 & -3404.000 & -1.000 & 11475600.000 & 10702176.000 \\ -2554.000 & -2188.000 & -1.000 & 0.000 & 0.000 & 0.000 & 1353620.000 & 1159640.000 \\ 0.000 & 0.000 & 0.000 & -2554.000 & -2188.000 & -1.000 & 4934328.000 & 4227216.000 \\ -2522.000 & -3288.000 & -1.000 & 0.000 & 0.000 & 0.000 & 1387100.000 & 1808400.000 \\ 0.000 & 0.000 & 0.000 & -2522.000 & -3288.000 & -1.000 & 7888816.000 & 10284864.000 \\ -3442.000 & -4316.000 & -1.000 & 0.000 & 0.000 & 0.000 & 5238724.000 & 6568952.000 \\ 0.000 & 0.000 & 0.000 & -3442.000 & -4316.000 & -1.000 & 13823072.000 & 17333056.000 \\ -3366.000 & -4400.000 & -1.000 & 0.000 & 0.000 & 0.000 & 4880700.000 & 6380000.000 \\ 0.000 & 0.000 & 0.000 & -3366.000 & -4400.000 & -1.000 & 13840992.000 & 18092800.000 \\ -3706.000 & -4368.000 & -1.000 & 0.000 & 0.000 & 0.000 & 6470676.000 & 7626528.000 \\ 0.000 & 0.000 & 0.000 & -3706.000 & -4368.000 & -1.000 & 14898120.000 & 17559360.000 \\ -3634.000 & -4464.000 & -1.000 & 0.000 & 0.000 & 0.000 & 6141460.000 & 7544160.000 \\ 0.000 & 0.000 & 0.000 & -3634.000 & -4464.000 & -1.000 & 14972080.000 & 18391680.000 \\ -3990.000 & -4436.000 & -1.000 & 0.000 & 0.000 & 0.000 & 7908180.000 & 8792152.000 \\ 0.000 & 0.000 & 0.000 & -3990.000 & -4436.000 & -1.000 & 16087680.000 & 17885952.000 \\ -3942.000 & -4540.000 & -1.000 & 0.000 & 0.000 & 0.000 & 7655364.000 & 8816680.000 \\ 0.000 & 0.000 & 0.000 & -3942.000 & -4540.000 & -1.000 & 16288344.000 & 18759280.000 \\ -3194.000 & -4600.000 & -1.000 & 0.000 & 0.000 & 0.000 & 4094708.000 & 5897200.000 \\ 0.000 & 0.000 & 0.000 & -3194.000 & -4600.000 & -1.000 & 13849184.000 & 19945600.000 \\ -3434.000 & -5576.000 & -1.000 & 0.000 & 0.000 & 0.000 & 5254020.000 & 8531280.000 \\ 0.000 & 0.000 & 0.000 & -3434.000 & -5576.000 & -1.000 & 17801856.000 & 28905984.000 \\ -2418.000 & -4160.000 & -1.000 & 0.000 & 0.000 & 0.000 & 1126788.000 & 1938560.000 \\ 0.000 & 0.000 & 0.000 & -2418.000 & -4160.000 & -1.000 & 9826752.000 & 16906240.000 \end{bmatrix}$$

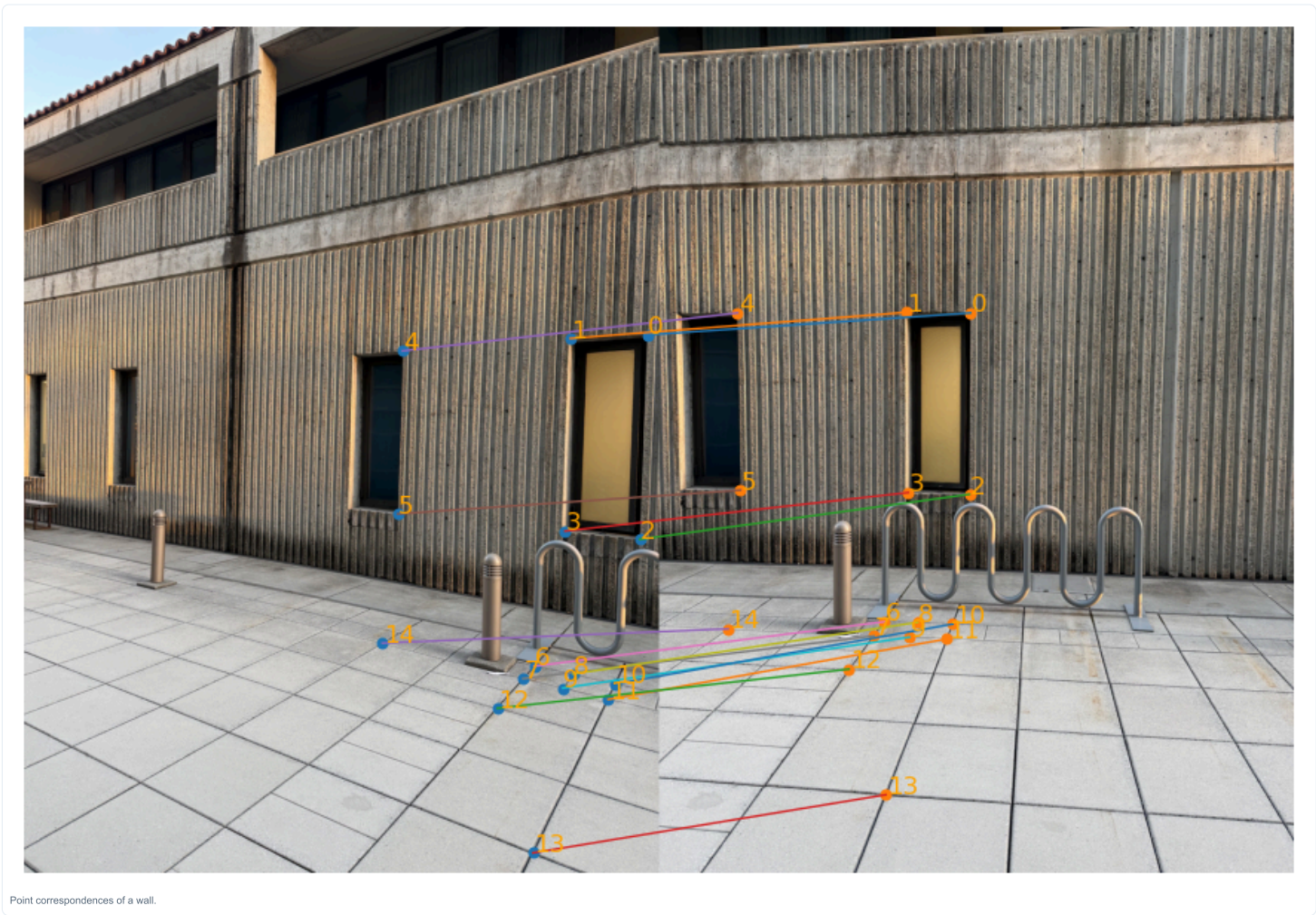
Vector $\mathbf{b}$ (shape 30×1):

-2102.000
-1932.000
-1670.000
-1928.000
-2094.000
-3156.000
-1682.000
-3144.000
-530.000
-1932.000
-550.000
-3128.000
-1522.000
-4016.000
-1450.000
-4112.000
-1746.000
-4020.000
-1690.000
-4120.000
-1982.000
-4032.000
-1942.000
-4132.000
-1282.000
-4336.000
-1530.000
-5184.000
-466.000
-4064.000

Recovered homography H

We report H normalized so that $H_{33} = 1$:

1.916	0.074	-4217.728
0.502	1.735	-2050.372
0.000	0.000	1.000



Point correspondences of a wall.

A.3: Warp the images

We rectify images via **inverse warping** using the same homography. The geometric mapping is identical; differences are purely in the *resampling* method.

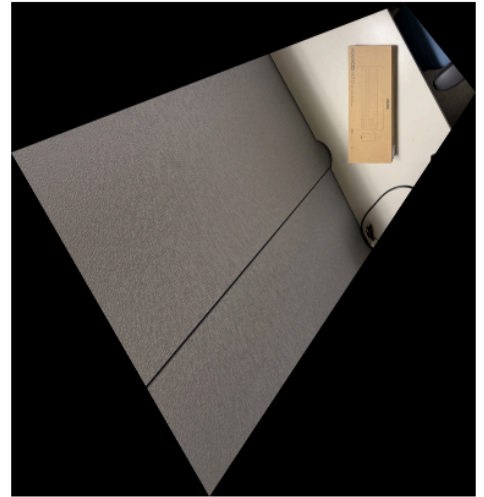
Original



Rectified (bilinear)



Rectified (nearest neighbor)

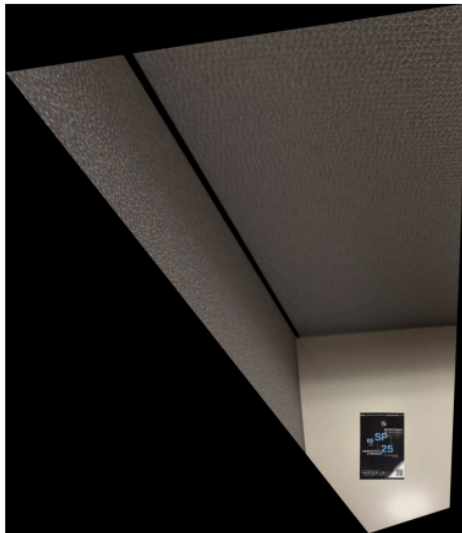


The rectified results of a rectangular box using both nearest neighbors and bilinear method.

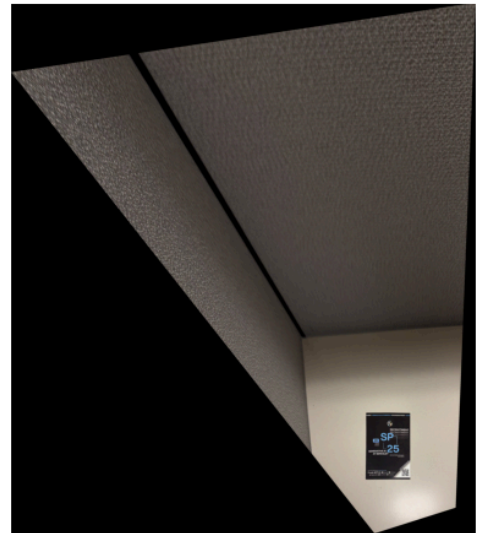
Original



Rectified (bilinear)



Rectified (nearest neighbor)



The rectified results of a rectangular flyer using both nearest neighbors and bilinear method.

Nearest Neighbor fastest

- **Look:** crisp but *jaggy*. Slanted edges (e.g., desk/partition seam and poster edges) show clear **stair-stepping** and aliasing. Fine textures can appear noisy or "sparkly".
- **Pros:** very fast; preserves discrete labels (useful for masks/segmentations).
- **Cons:** visible artifacts on diagonals/thin lines; can produce moiré on repetitive patterns.

Bilinear better quality

- **Look:** visually **smoother**. The same slanted edges become continuous with much less stair-stepping; repetitive textures exhibit far less aliasing. Small text/high-frequency detail is *slightly softened* (minor blur).
- **Pros:** reduces aliasing and edge artifacts; usually preferred for perceptual quality and for blending in mosaics.
- **Cons:** mild blur; a bit slower than NN (still cheap in practice).

Conclusions

- For **rectification** deliverables, *bilinear* yields cleaner edges and fewer artifacts; NN is great for quick debugging.
- For **mosaics**, bilinear generally produces better seams and fewer high-frequency artifacts, especially when combined with feathering or pyramid blending.
- Both methods avoid holes thanks to **inverse warping**: geometry is identical—the difference is only photometric (resampling).

A.4: Blend images into a mosaic

We form a mosaic by warping one image into the coordinate system of the other and combining them with *soft alpha feathering* (weighted averaging). The geometry comes from the homography; the visual smoothness comes from our weights.

Inputs & Outputs

- **Inputs:** `im1` (to be warped), `im2` (reference), `H_1to2` (homography mapping `im1` $\rightarrow$ `im2`), a warper `warp_func` (either `warpImageNearestNeighbor` Or `warpImageBilinear`), and blending parameters `feather`, `falloff`, `gamma`, `bg`.
- **Output:** the blended mosaic image.

Procedure (what `blend_pair` does)

1. **Warp `im1` into `im2`'s frame** (inverse warping). We call `warp_func(im1, H_1to2, fill_value=bg)` which returns: the warped image `w1`, a binary alpha `a1` (0 outside, 255 where samples are valid), and the integer `offset` of the warped patch within the mosaic canvas.
2. **Compute the mosaic canvas bounds.** We take the union of the reference image extent and the warped patch (using the returned offset and size) to get the mosaic width `w` and height `h`.
3. **Initialize accumulators.** Two arrays of shape `(h, w, 3)`:
 - `canvas_sum`: accumulates `pixel * weight`

◦ `canvas_wsum`: accumulates weight

These let us do a stable weighted average at the end.

4. **Build soft alpha weights (feathering).** For each image we create a smooth weight map:

- Start with a binary mask (all 255 for `im2`; the returned `a1` for `w1`).
- Run a separable **box blur** with odd kernel size `ksize = falloff`. We implement the blur with an *integral-image* style 1-D pass that preserves the original array size.
- Normalize to `[0,1]`, zero outside original support, and optionally apply `gamma` to sharpen or widen the ramp.

The helper is `_soft_alpha(alpha_uint8, ksize, gamma)`.

5. **Accumulate both layers.** Paste `im2` (unwarped) at its natural location and `w1` at its offset:

```
canvas_sum += patch * weight
canvas_wsum += weight
```

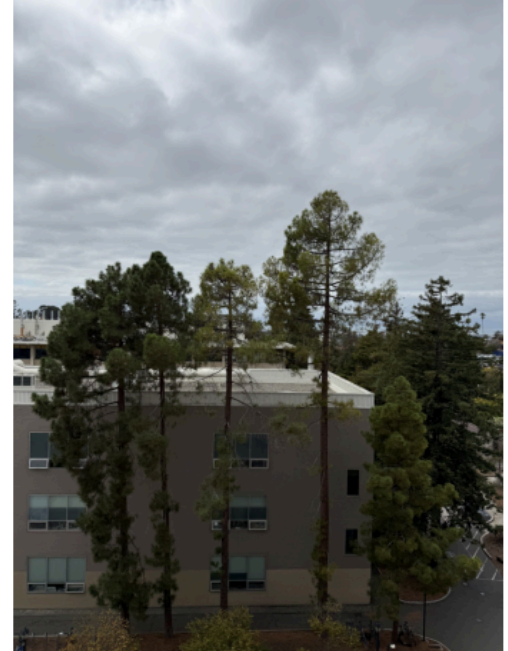
We use `_accumulate(...)` to handle offsets, channels, and broadcasting.

6. **Normalize to get the mosaic.** `mosaic = canvas_sum / max(canvas_wsum,  $\epsilon$ )` with a small ϵ to avoid division by zero.

im1



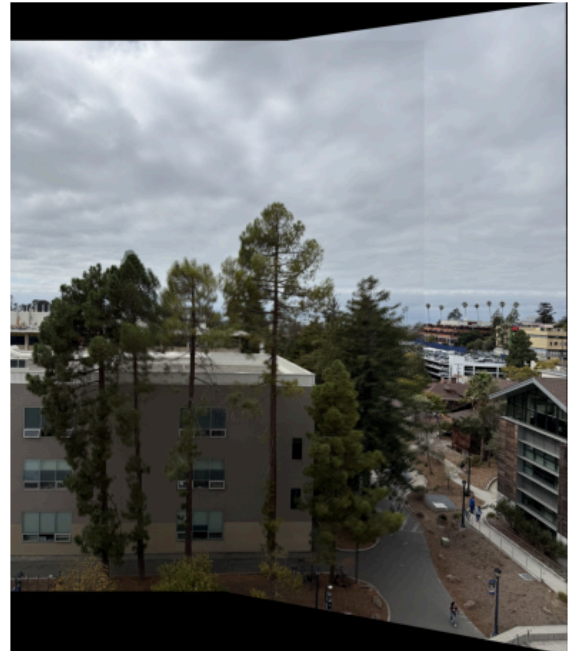
im2



Mosaic (NN)



Mosaic (Bilinear)



Building 1 mosaic.

im1



im2



Mosaic (NN)



Mosaic (Bilinear)

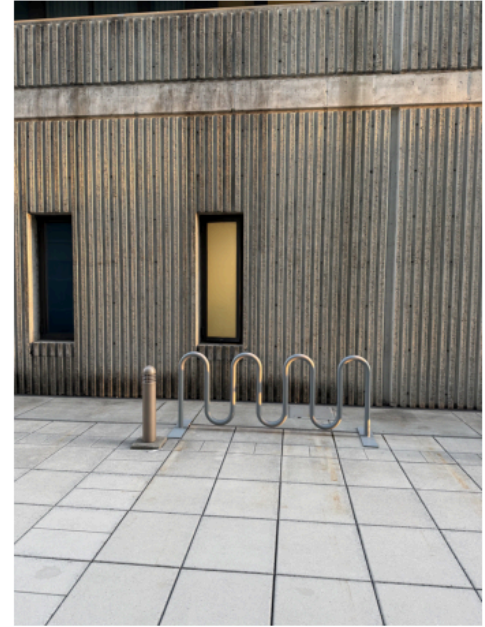


Building 2 mosaic.

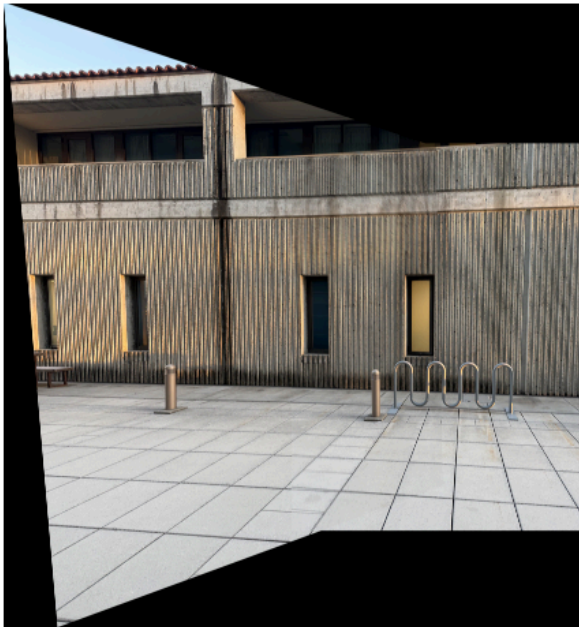
im1



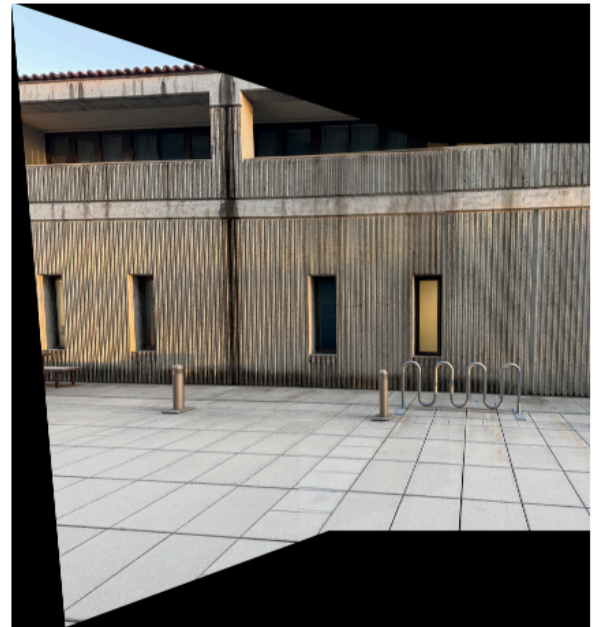
im2



Mosaic (NN)



Mosaic (Bilinear)

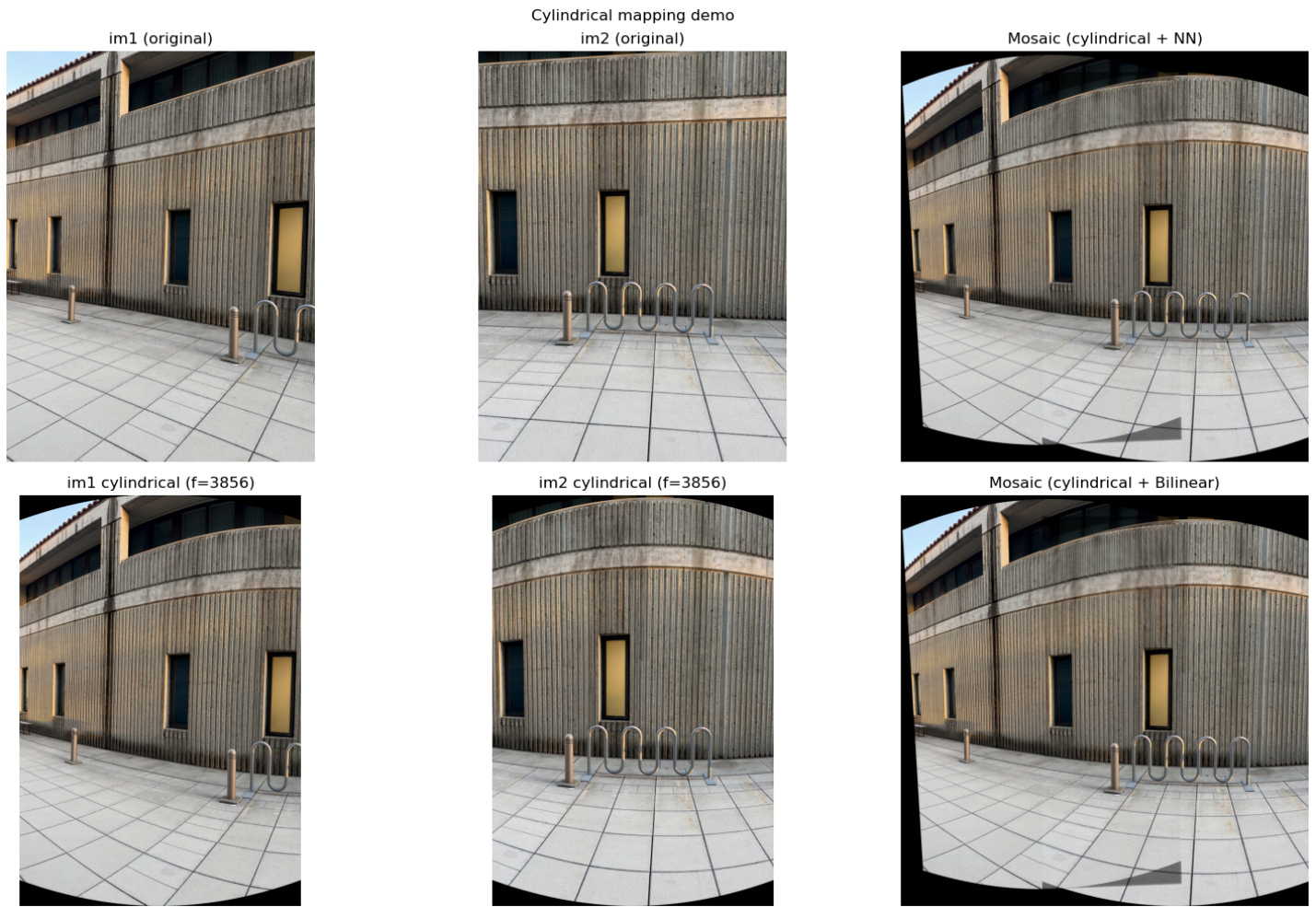


The wall mosaic.

A.5: Bells and Whistles

Instead of projecting each image onto a plane, we project it onto the surface of a cylinder and then "unroll" the cylinder to a flat image. With focal length f (in pixels) and principal point (c_x, c_y) , the horizontal coordinate is reparameterized by an angle, which reduces perspective distortion for wide fields of view:

$$x' = f \arctan\left(\frac{x - c_x}{f}\right), \quad y' = f \frac{(y - c_y)}{\sqrt{(x - c_x)^2 + f^2}} + c_y.$$



The cylindrical mapping of the wall.

B.1: Harris Corner Detection

See the figures below. If we directly use the Harris interest point detector, there will be large amount of points. Using ANMS will make the interest point more robust.



The comparison of direct harris algorithm and anms algorithm (100 points) for the building 1.

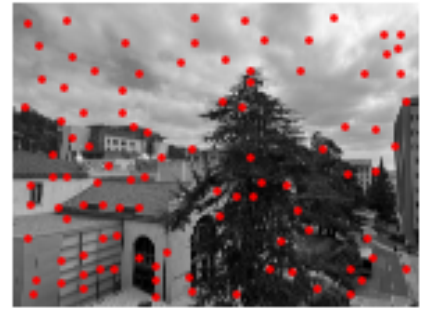
Original Image



Harris Corners



ANMS Corners

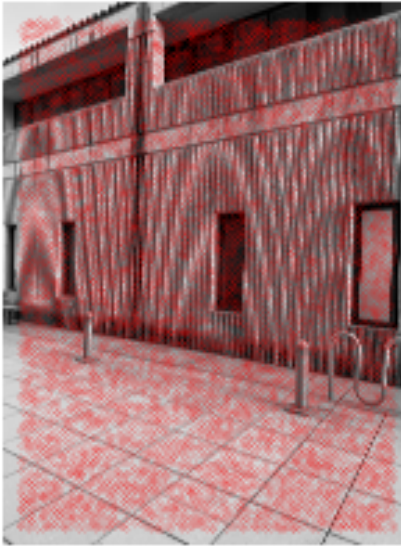


The comparison of direct harris algorithm and anms algorithm (100 points) for the building 2.

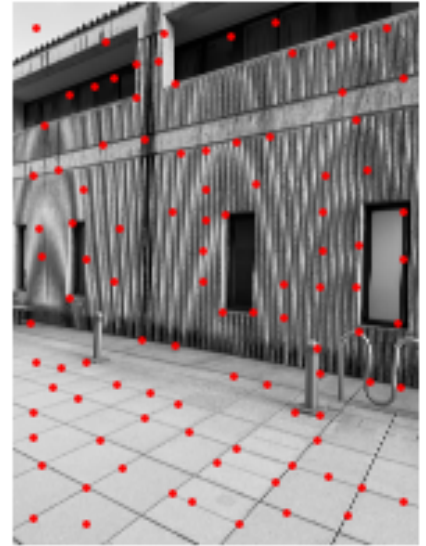
Original Image



Harris Corners



ANMS Corners

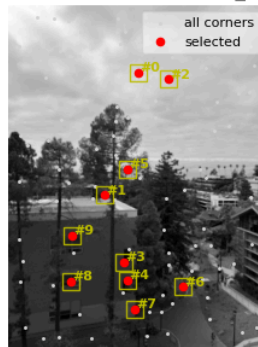


The comparison of direct harris algorithm and anms algorithm (100 points) for the wall.

B.2: Feature Descriptor Extraction

For each place, we select 10 representative feature descriptors.

Keypoints and sampling windows (patch_size=8, spacing=5)



#0 idx=49
(y=140, x=269)



#1 idx=97
(y=393, x=199)



#2 idx=53
(y=153, x=331)



#3 idx=5
(y=531, x=239)



#4 idx=33
(y=569, x=248)



#5 idx=65
(y=339, x=248)



#6 idx=62
(y=582, x=361)



#7 idx=51
(y=629, x=261)



#8 idx=38
(y=571, x=130)

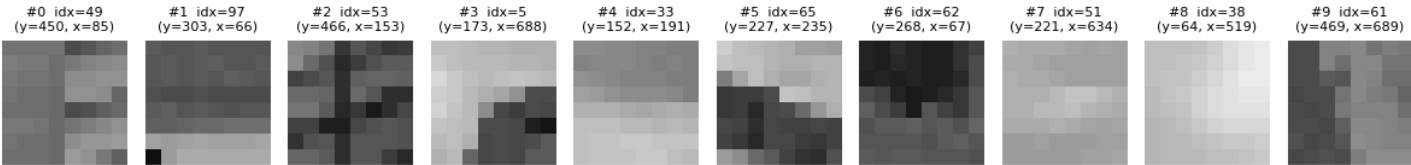
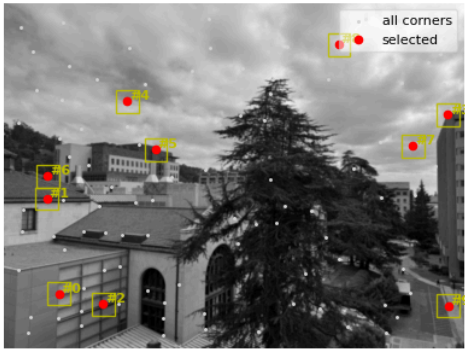


#9 idx=61
(y=476, x=132)



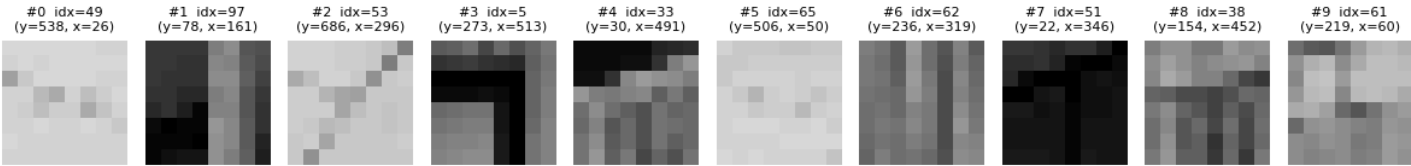
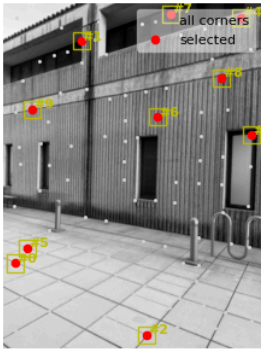
Representative feature descriptors of the building 1.

Keypoints and sampling windows (patch_size=8, spacing=5)



Representative feature descriptors of the building 2.

Keypoints and sampling windows (patch_size=8, spacing=5)

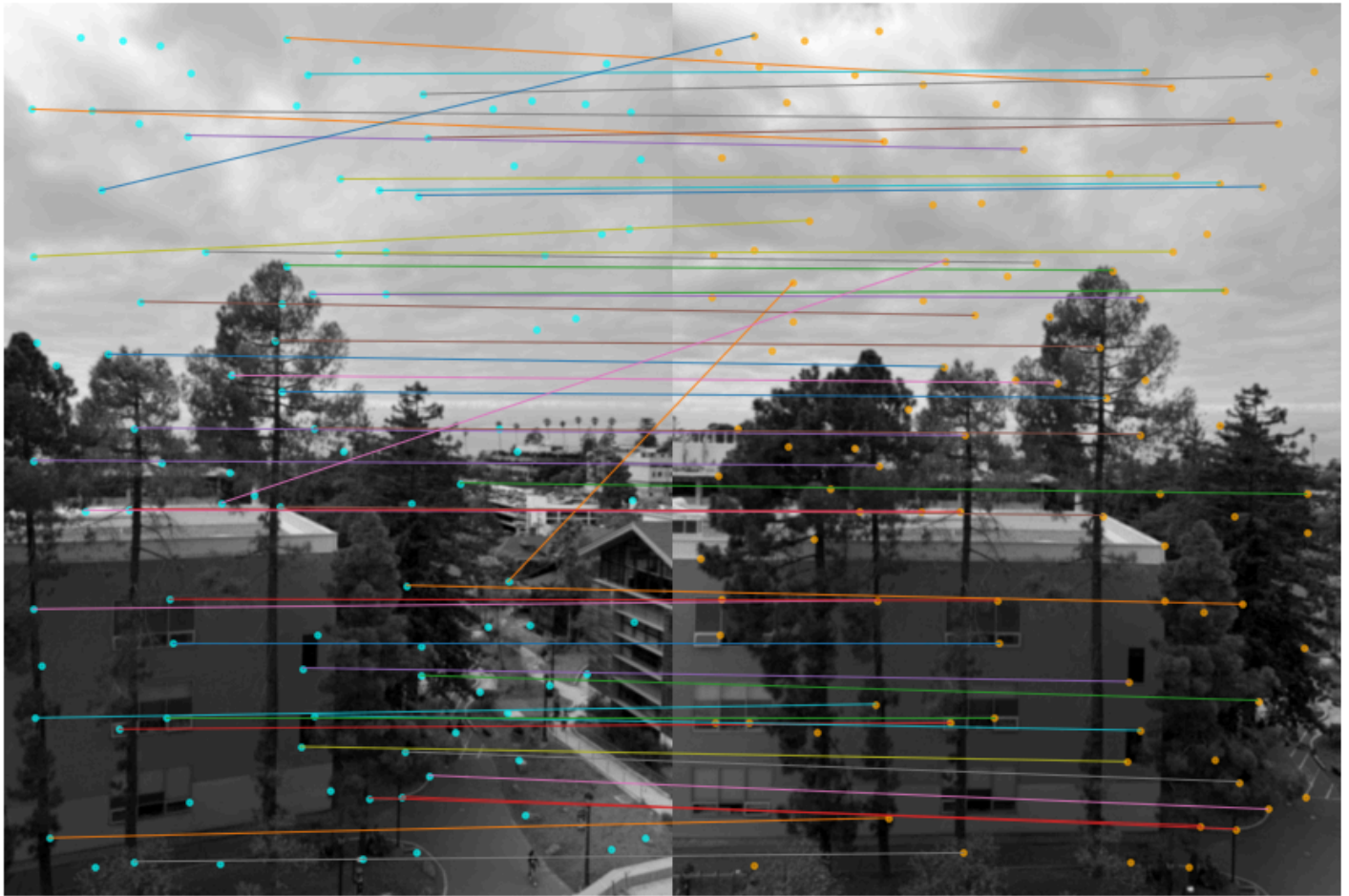


Representative feature descriptors of the wall.

B.3: Feature Matching

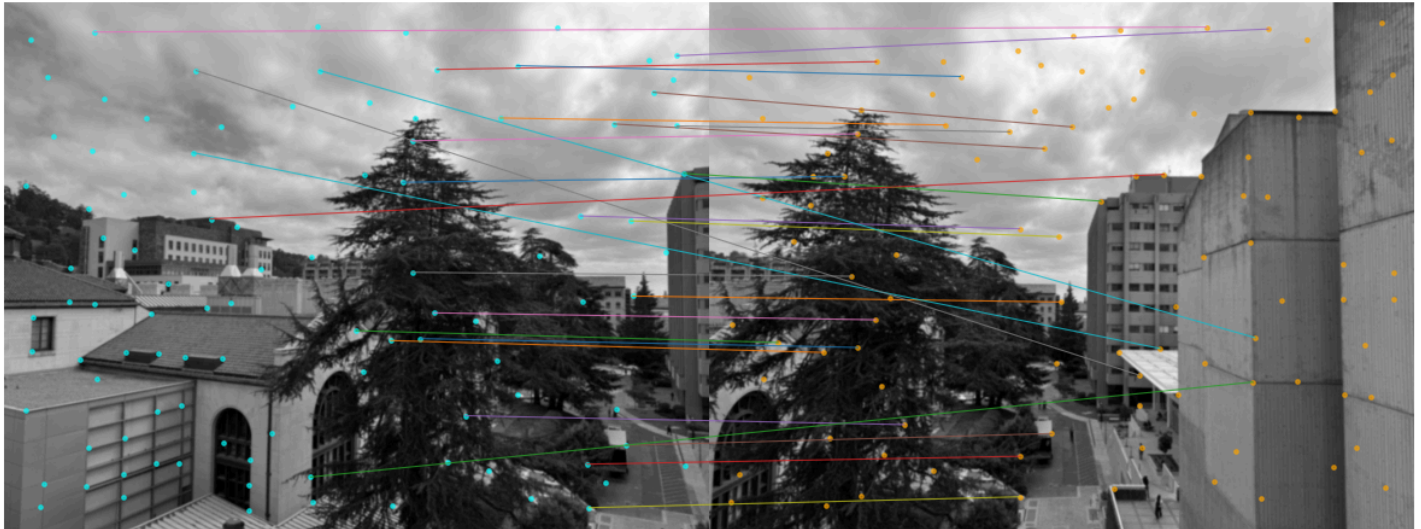
For each place, we choose 0.8 as the ratio threshold.

Matched features: 48



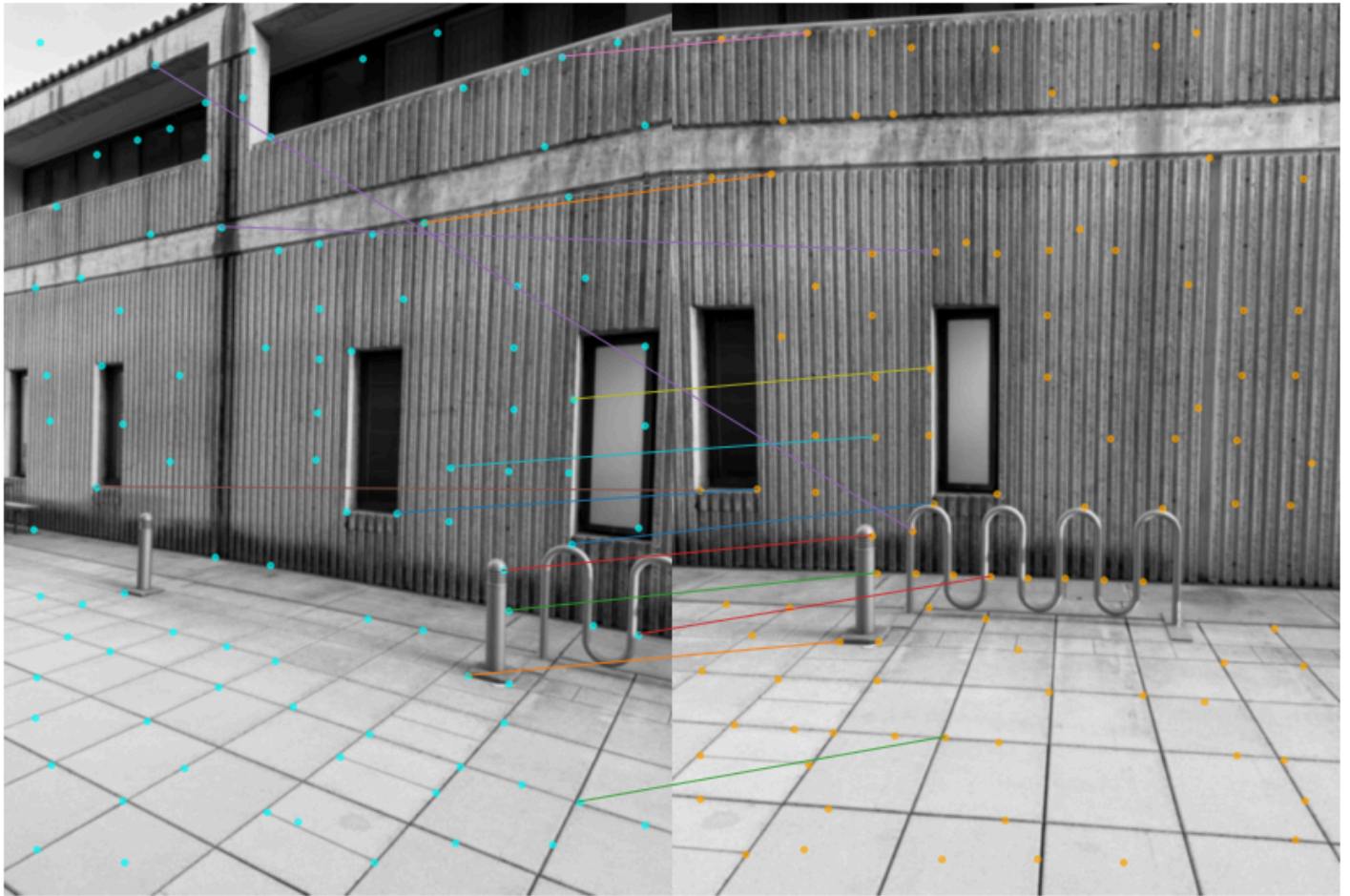
Feature matching of the building 1.

Matched features: 28



Feature matching of the building 2.

Matched features: 15



Feature matching of the wall.

B.4: RANSAC for Robust Homography

RANSAC Homography Stitch (auto)



Auto stitching using RANSAC of the building 1.

RANSAC Homography Stitch (auto)



Auto stitching using RANSAC of the building 2.

RANSAC Homography Stitch (auto)



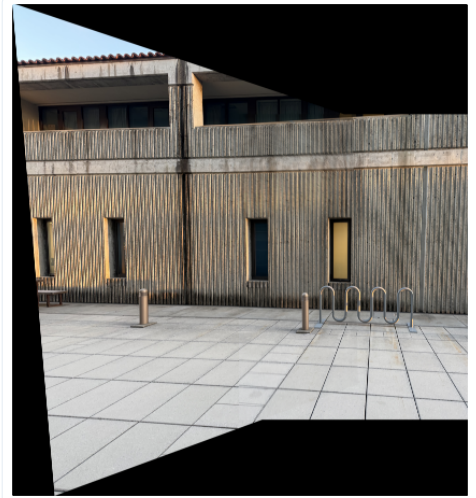
Auto stitching using RANSAC of the wall.



Manually stitching using RANSAC of the building 1.



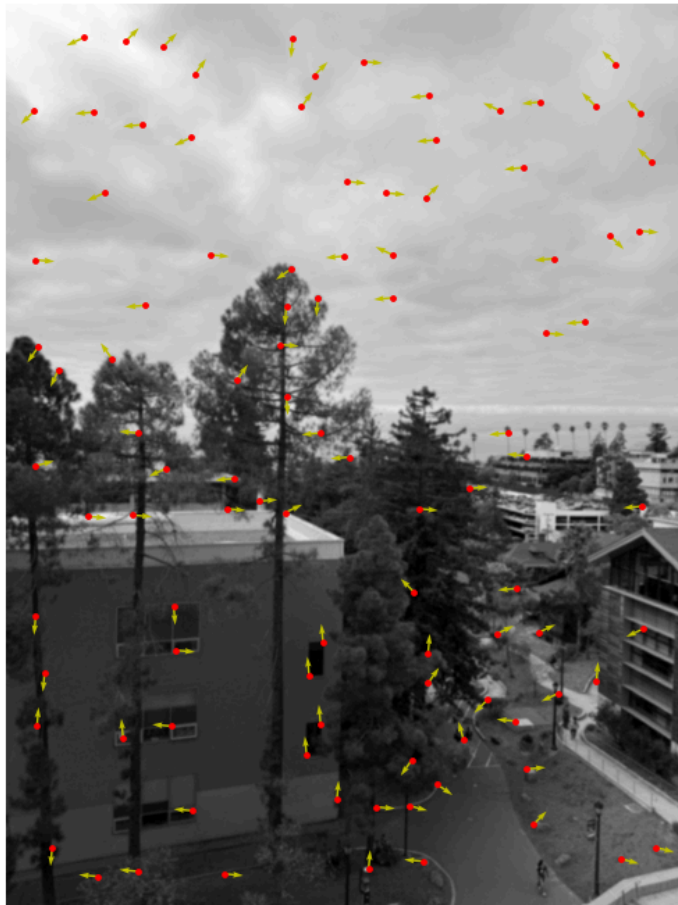
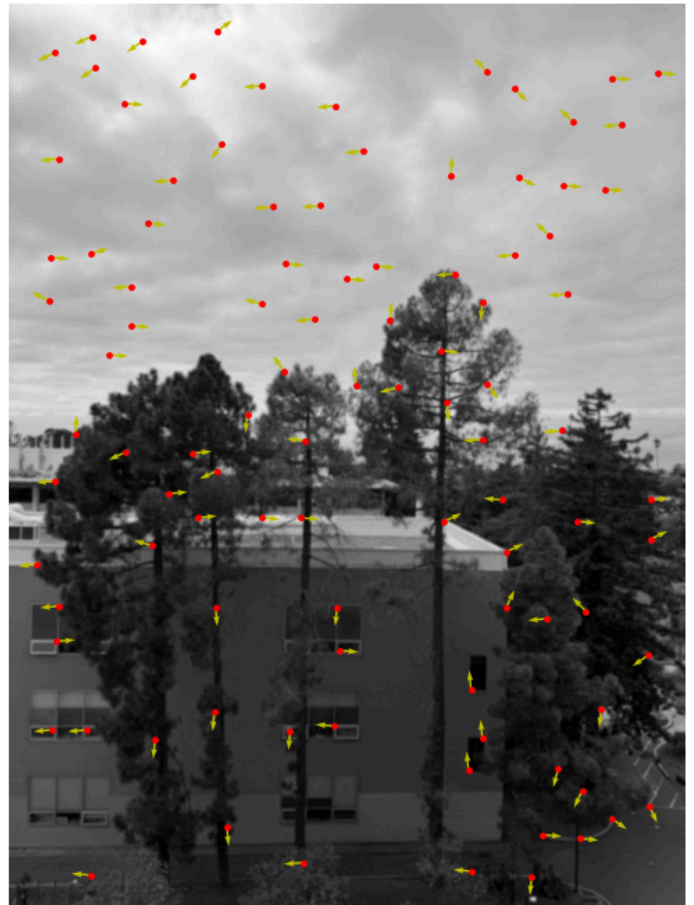
Manually stitching using RANSAC of the building 2.



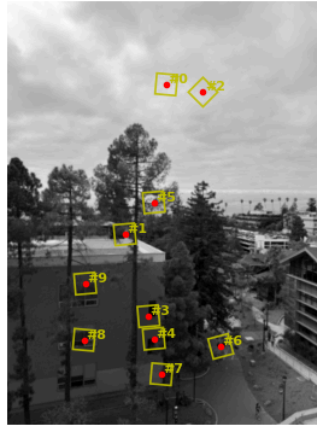
Manually stitching using RANSAC of the wall.

B.5: Rotation invariance

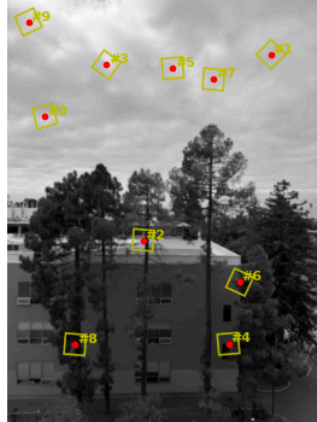
For the building 1, the feature descriptors with rotation invariance is shown below.

Keypoint orientations**Keypoint orientations**

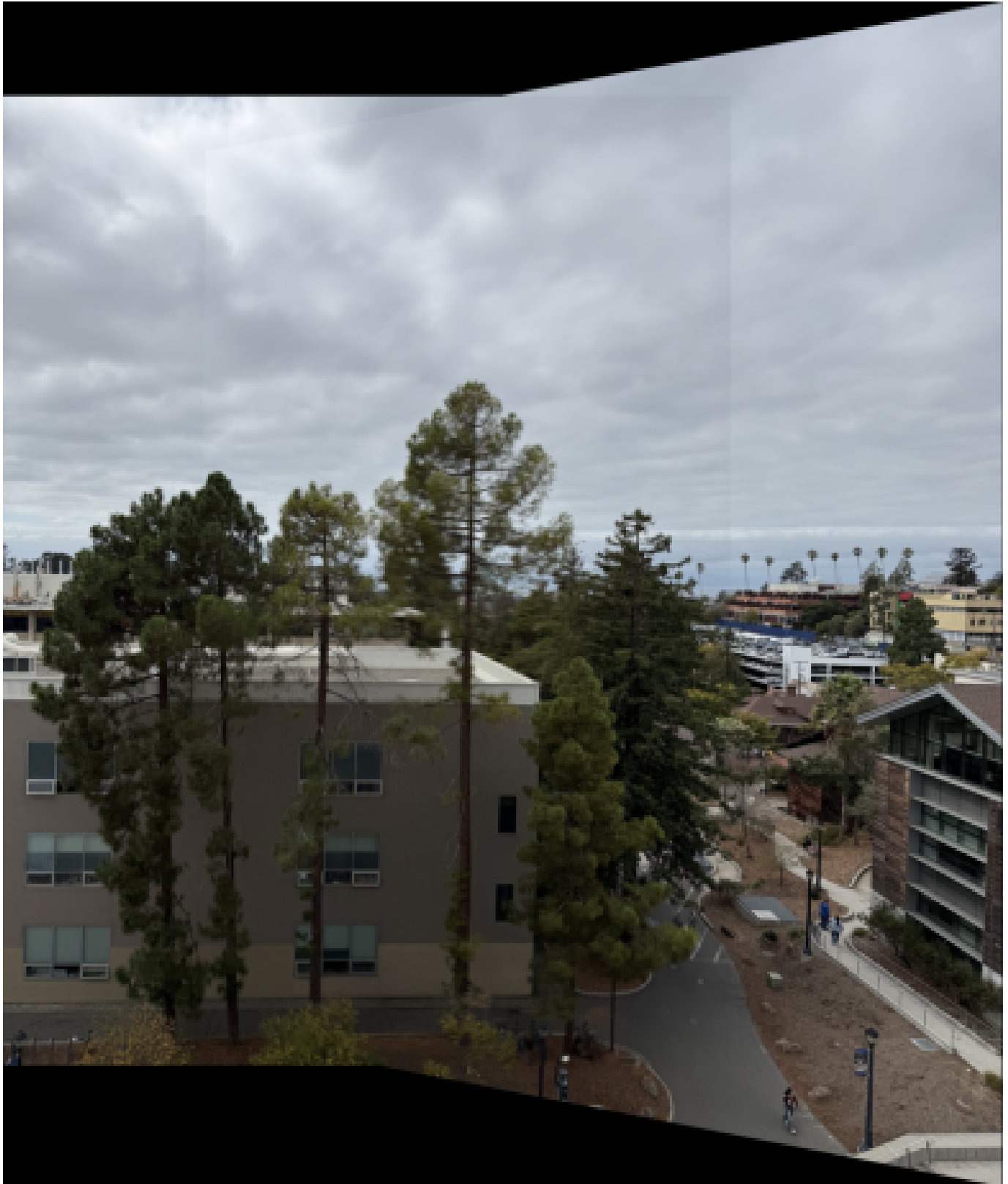
Rotated sampling windows

#0 $\theta=5^\circ$ #1 $\theta=-5^\circ$ #2 $\theta=-45^\circ$ #3 $\theta=-95^\circ$ #4 $\theta=-95^\circ$ #5 $\theta=175^\circ$ #6 $\theta=-105^\circ$ #7 $\theta=-85^\circ$ #8 $\theta=-175^\circ$ #9 $\theta=85^\circ$ 

Rotated sampling windows

#0 $\theta=-15^\circ$ #1 $\theta=-135^\circ$ #2 $\theta=5^\circ$ #3 $\theta=125^\circ$ #4 $\theta=-95^\circ$ #5 $\theta=175^\circ$ #6 $\theta=-65^\circ$ #7 $\theta=-85^\circ$ #8 $\theta=95^\circ$ #9 $\theta=155^\circ$ 

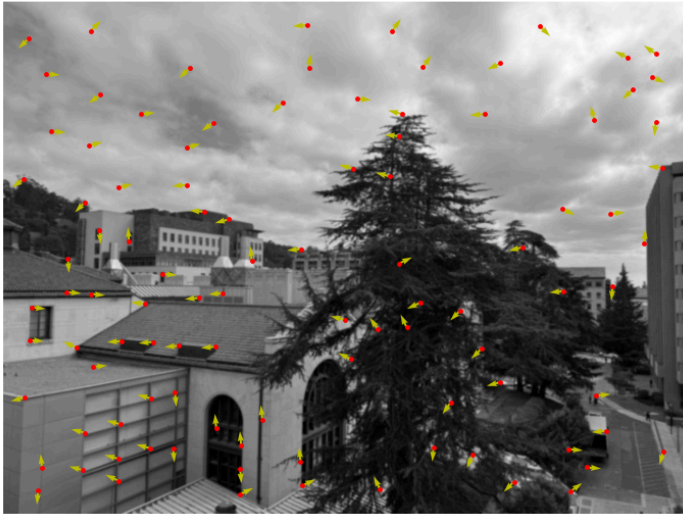
Auto stitch with rotation-invariant descriptors



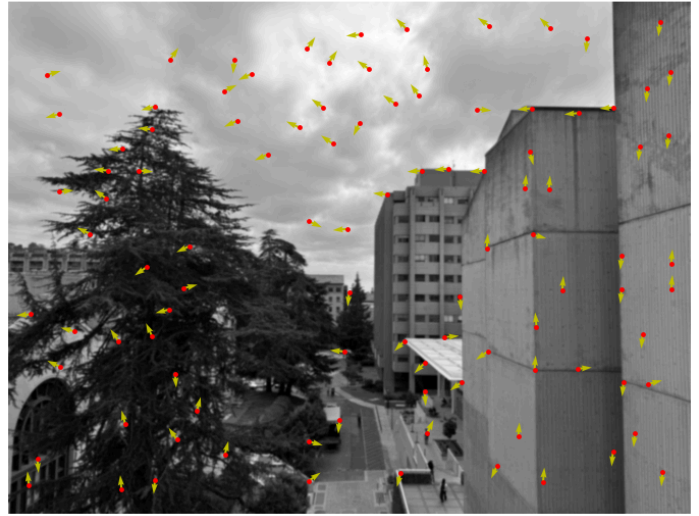
Auto stitching with rotation invariance descriptors of the building 1.

For the building 2, the feature descriptors with rotation invariance is shown below.

Keypoint orientations



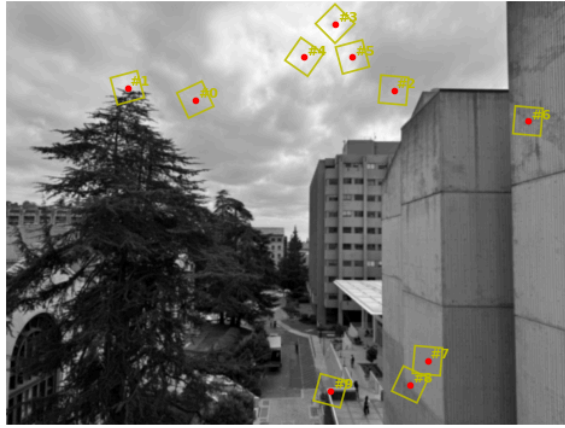
Keypoint orientations



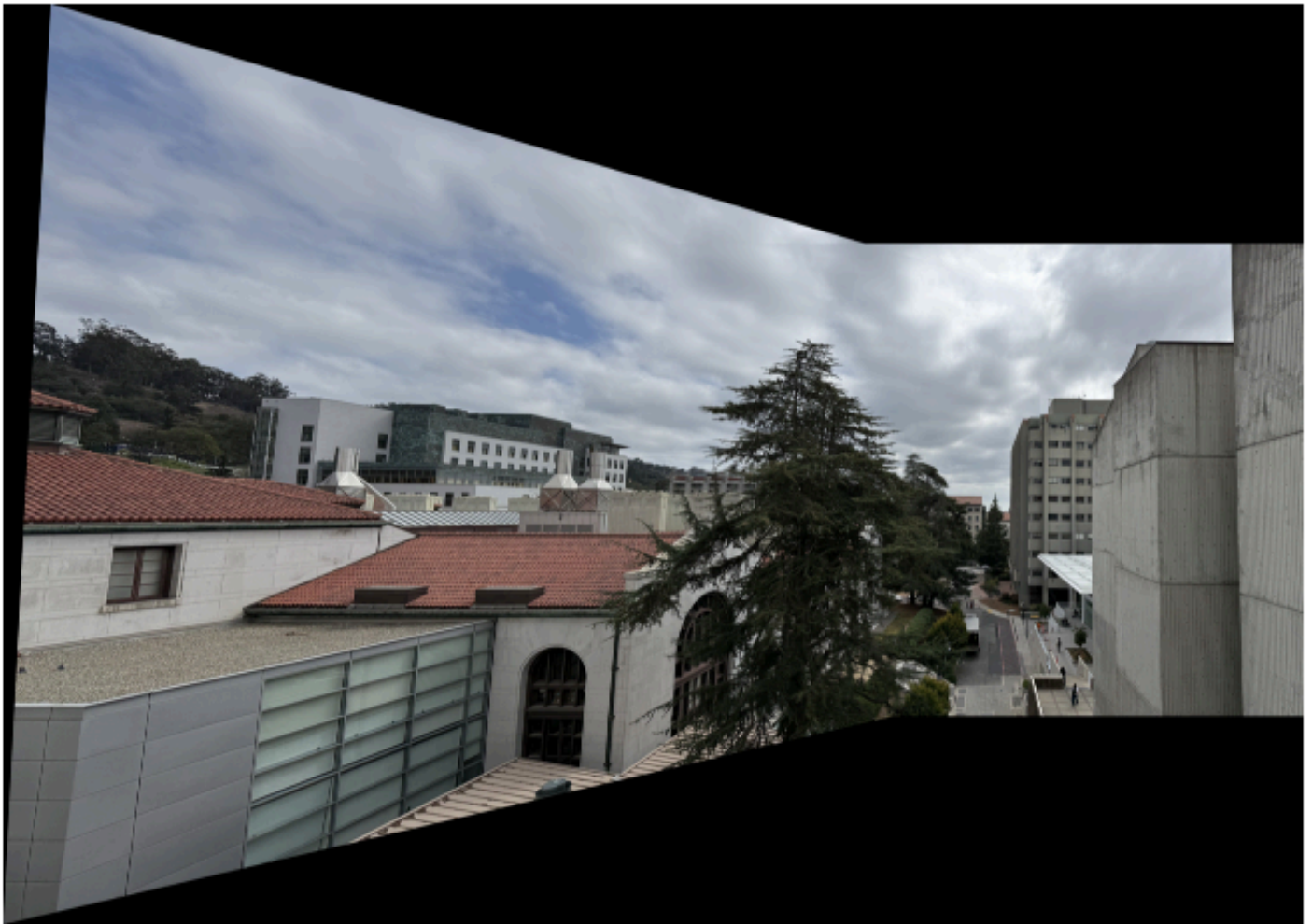
Rotated sampling windows

#0 $\theta = -165^\circ$ #1 $\theta = 5^\circ$ #2 $\theta = -165^\circ$ #3 $\theta = -175^\circ$ #4 $\theta = -15^\circ$ #5 $\theta = 165^\circ$ #6 $\theta = 85^\circ$ #7 $\theta = -5^\circ$ #8 $\theta = 155^\circ$ #9 $\theta = 105^\circ$ 

Rotated sampling windows



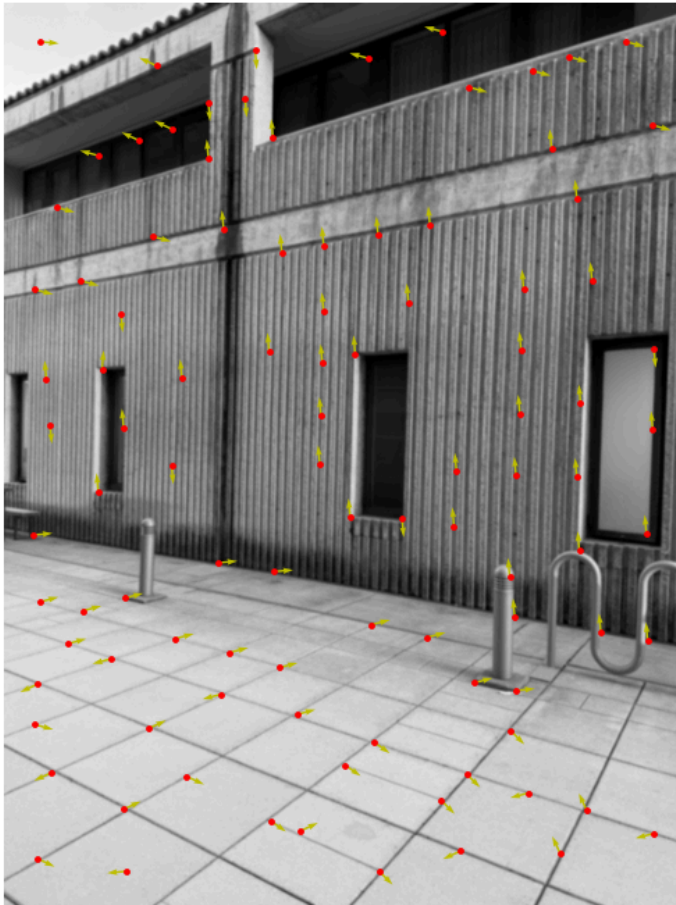
Auto stitch with rotation-invariant descriptors



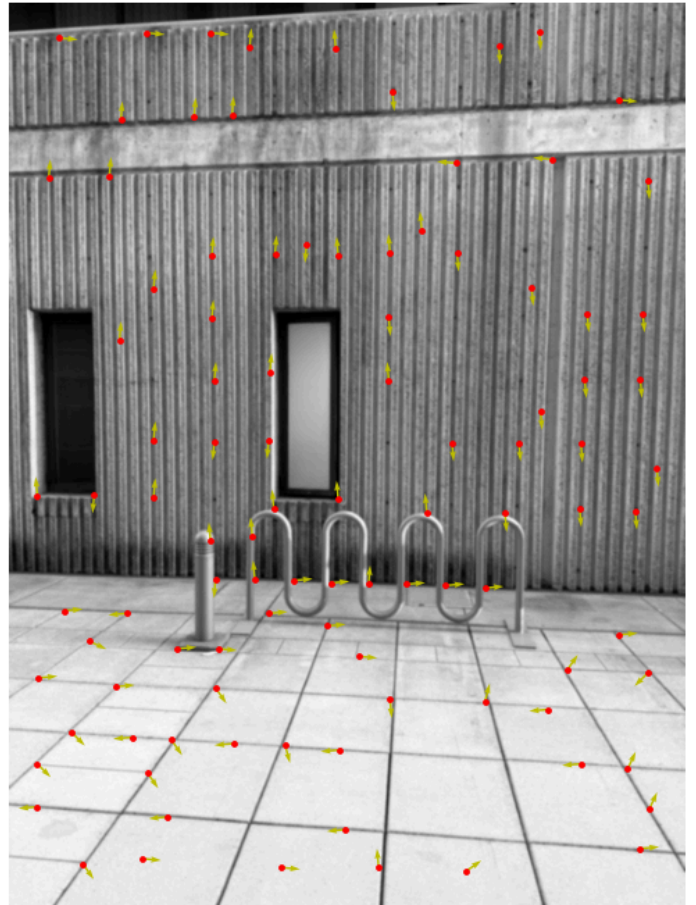
Auto stitching with rotation invariance descriptors of the building 2.

For the wall, the feature descriptors with rotation invariance is shown below.

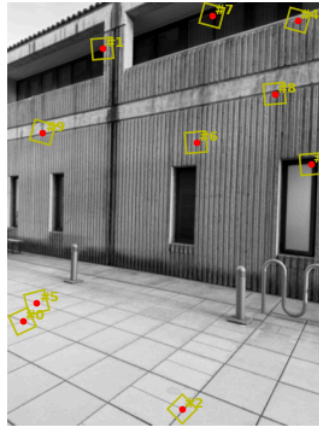
Keypoint orientations



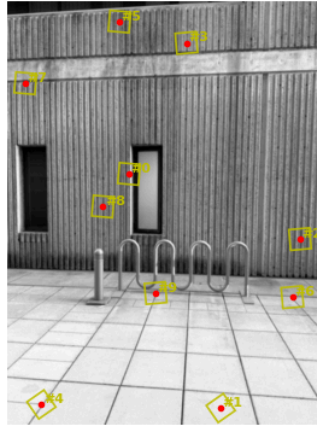
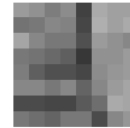
Keypoint orientations



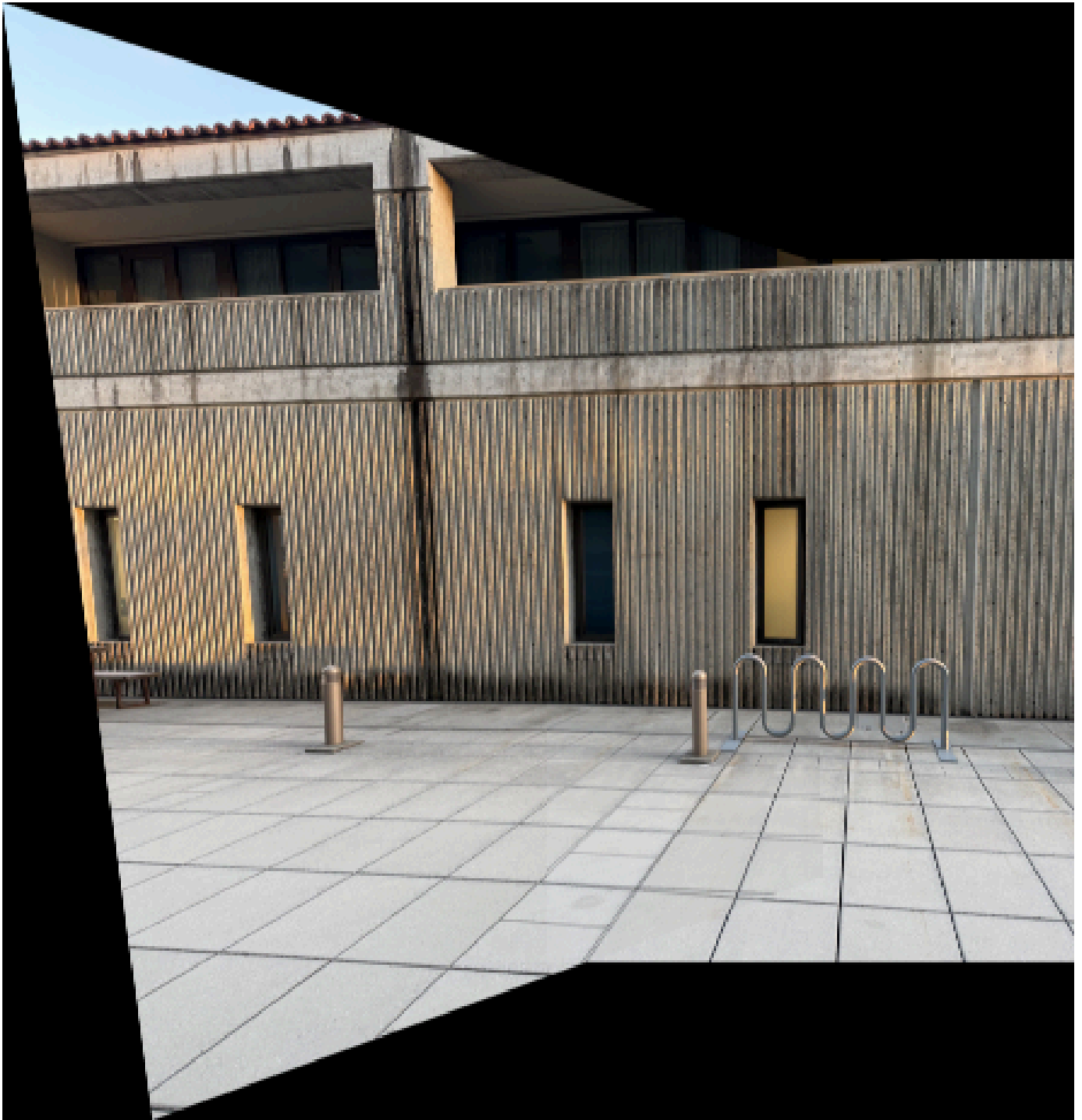
Rotated sampling windows

#0 $\theta=155^\circ$ #1 $\theta=85^\circ$ #2 $\theta=45^\circ$ #3 $\theta=85^\circ$ #4 $\theta=15^\circ$ #5 $\theta=-15^\circ$ #6 $\theta=-95^\circ$ #7 $\theta=-165^\circ$ #8 $\theta=-95^\circ$ #9 $\theta=15^\circ$ 

Rotated sampling windows

#0 $\theta = -85^\circ$ #1 $\theta = -35^\circ$ #2 $\theta = 85^\circ$ #3 $\theta = 85^\circ$ #4 $\theta = 55^\circ$ #5 $\theta = -85^\circ$ #6 $\theta = -5^\circ$ #7 $\theta = -85^\circ$ #8 $\theta = 95^\circ$ #9 $\theta = -5^\circ$ 

Auto stitch with rotation-invariant descriptors



Auto stitching with rotation invariance descriptors of the wall.